

Combining Local Lesion and Global Mammographic Context for Breast Mass Classification

A Study Using the CBIS-DDSM Dataset

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August 2026

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1 Overview of the project

According to the World Health Organization, breast cancer is one of the leading causes of death among women. It was the most common cancer in women in 157 countries out of 185 in 2022. 670,000 women died from this disease worldwide in the same year. Statistically, over the course of their lives, 1 woman out of 12 will suffer from breast cancer, making a compelling case for prevention. Although the techniques may vary, one of the most prevalent is mammography. It is a well-established imaging technique that has undergone significant improvements ranging from digital mammography to digital breast tomosynthesis (Wang 2024). Historically, mammograms were interpreted exclusively by specialists, but CNN-based methods are increasingly used to support diagnostic workflows.

Relating to the template 3.2. “Deep Learning Breast Cancer Detection”, the project aims to classify suspicious mammographic abnormalities as benign or malignant. Mammography assessment can be divided into two main tasks: identifying abnormalities in the image, known as computer-aided detection (CADe), and assessing whether an identified abnormality is benign or malignant, known as computer-aided diagnosis (CADx). While CADe may prompt further investigation, CADx supports the diagnostic characterisation of an identified abnor-

mality. This project focuses on the latter.

Traditionally, this assessment is performed by radiologists, usually two of them for one patient. Their assessments provide an early indication of the nature of the tumour, but they are subject to inter-reader variability, fatigue, workload and differences in expertise. Computer-aided diagnosis has developed partly to address these inherent limitations.

Originally, CAD relied on traditional machine-learning approaches. Tahmasbi et al. (Tahmasbi et al. 2011) achieved high accuracy by using Zernike moments without requiring a large dataset. In a field where data is scarce and of variable quality, this is an upside. However, traditional machine learning models require manual feature selection, which is also human-dependent and prone to biases. Deep learning reduces this dependence on manual feature selection by learning visual representations directly from images, but requires sufficient representative data and computational resources.

This project consists of developing a CNN-based model. CNNs are particularly well suited to computer-vision and image analysis tasks. However, in the case of CADx on mammograms, they also come with challenges. Curated datasets remain relatively limited, while annotations and acquisition artefacts may introduce misleading visual patterns. Medical predictions also require careful interpretation and appropriate evaluation: in particular, sensitivity is important because missing a malignant lesion may have more serious consequences than incorrectly flagging a benign case.

This project therefore aims to develop a reproducible CADx pipeline that reduces the influence of non-anatomical annotations and classifies known mammographic masses using lesion-level and breast-level tissue information. This objective involves a central methodological tension: curated mammography datasets are relatively limited in size, whereas CNNs generally require sufficient and representative data to learn robust visual features. The project addresses this constraint through controlled preprocessing, transfer learning, regularisation and a local-global architecture.

- Research question: Does combining an ROI-centred lesion representation with contextual information from the full breast improve benign-malignant classification compared with either representation alone?
- Hypothesis: A model combining local lesion features with global breast context will achieve better validation performance than the best single-branch model.

2 Literature review

2.1 CBIS-DDSM Dataset

Varoquaux and Cheplygina emphasise that medical datasets suffer from several biases. Although it is a common practice in machine learning to build a test set that is a subset of the same data pool as the train set, there is a risk of validating a model’s performance on a population that does not reflect the distribution of the actual population, causing poor performance of the model in real-life situations. In addition, a dataset may not necessarily capture the full extent of medical conditions, leading to model underperformance if it was trained on data underestimating them. The dataset also requires high-quality and curated images. Indeed, they can sometimes be affected by traces of medical intervention, leading the model to learn inappropriate features. Besides, the labelling needs to be accurate and of high quality. Finally, the set should not contain any patient leakage because it can overestimate the obtained result by allowing the model to learn patterns that will be found in the test set (Varoquaux and Cheplygina 2022).

Being aware of these shortcomings, this study uses the CBIS-DDSM reference dataset. It is a curated version of the DDSM dataset, which is a collection of mammograms from “Massachusetts General Hospital, Wake Forest University School of Medicine, Sacred Heart Hospital, and Washington University of St Louis School of Medicine” (Lee et al. 2017). To some extent, the CBIS-DDSM dataset represents an attempt to tackle the biases the legacy DDSM faced as described by Lee et al. First, the accessibility of data is improved. The meta-data are provided in two CSV files, including patient characteristics (BI-RADS, age, etc). Some mass cases were also questionable as to whether they represented genuine signs of disease. With the help of a trained mammographer, 339 images were removed that could not be appropriately confirmed. Considering the original DDSM dataset accounted for 2620 scanned film mammography studies, this removal represents a substantial proportion of the original dataset and reduces the risk of training on uncertain cases. The CBIS-DDSM dataset also comes with an enhanced region-of-interest (ROI) segmentation for mass cases. The original DDSM provided a mere location of the mass tumour, which is not sufficient for a model to properly train. By using the original ROI as a rough starting point, the region was further cropped to the exact contour using Chan-Vese model (Chan and Vese 2001). Finally, the CBIS-DDSM also comes up with a standardized train/test split. For mass cases, on which this study focuses, the train set accounts for 51.37% benign cases and 48.63% malignant cases. The test set, on the other hand, is 58.5% benign and 41.5% malignant.

2.2 Critical assessment of literature

2.2.1 Architectures

2.2.1.1 Input The CBIS-DDSM is a bank of medical cases. A case indicates a certain abnormality. The medical process entails two mammograms per abnormality, the craniocaudal (CC) and the mediolateral oblique (MLO) views. The CBIS-DDSM dataset also comes with a cropped image of the region and a mask for the full image. Every patient examination may thus come up with four documents. This structure therefore enables several preprocessing strategies. Liao and Aagaard convert all DICOMs into PNG. The authors built a pipeline that consists in applying the mask to the full mammogram and that centres around the tumour. The mask is strict, meaning that pixels outside the region of interest are set to black. Ruling out the default cropped images allows the authors to control the preprocessing pipeline rather than relying directly on the provided cropped images. As a prerequisite, the mask dimensions should match those of the mammogram. Otherwise, the case is excluded because it is not considered reliable (Liao and Aagaard 2024). The main limitation of a strict ROI-masking approach is that it may remove contextual information surrounding the lesion.

In contrast to the lesion-focused approach of Liao et al., Abdikenov et al. emphasise that tumour diagnosis occurs in a contextual analysis of the full mammogram. The presence of calcifications - compatible with a malignant form - the breast density, whether it is a CC or a MLO view orient the radiologist’s judgment. Their study concludes that training two models, one on CC views and the other one on MLO views, and, then, merging their features, gives better results than training a model on a merged image of both CC and MLO views (Abdikenov et al. 2025). As such, it emphasises the importance of the context. The architecture can somehow perform well under some models, like VGG19 (accuracy : 0.8865 and F1 : 0.6783) or DenseNet (accuracy : 0.7152 and F1 : 0.7582), but direct image merging may blur the signal and prevent generalisation. According to the authors’ findings, two distinct models generalize better on heterogeneous data when trained on well-defined views.

The approach does not consider any crop though, meaning the tumour assessment is achieved in the context of the entire mammogram. The mammogram is considered as a full input and, as such, there is no notion of region of interest. A full-image or multi-view approach may capture contextual information, but it may also make it harder to isolate the contribution of the lesion itself because the patterns are learnt from an image containing a lot more information than the presence of a mass. There is a risk that the model learns global correlation associated with malignancy rather than lesion-specific features. Furthermore, depending on whether the mammograms were preprocessed, they may display human artifacts or traces of handwriting, which could introduce shortcut learning or spurious correlations. Shen, et al. therefore adopt a hybrid approach combining a full mammogram view and ROI study called GMIC (Shen et al.

2021). Although the model trained at the full mammogram level has a CAdE purpose (detection of an abnormality), its features are combined with those produced by the local module, encouraging the CAdx process to acknowledge the full context. This study reports performance comparable to radiologists under the conditions evaluated while requiring less GPU memory and being 4.3 times faster than ResNet-34.

Source	Main contribution	Relevance to this project	Limitation / gap
Liao et al.	Propose a transparent CBIS-DDSM pipeline using ROI masks applied to full mammograms. Pixels outside the ROI are set to black, creating a strict lesion-focused representation.	Supports the use of hard ROI masking as a controlled preprocessing strategy for CAdx. It helps the CNN focus on the annotated abnormality rather than unrelated image regions.	Strict masking may remove useful surrounding context, such as lesion margins, nearby tissue patterns, or breast density information.
Abdikenov et al.	Compares multi-view mammography strategies using CC and MLO views. Their results suggest that processing views separately before feature fusion can outperform simple image merging.	Supports the idea that mammographic context and view-specific information matter for classification. It motivates looking beyond isolated lesion crops.	Their approach uses full-view information and does not directly test ROI masking or lesion-centred representations. It therefore does not isolate the effect of preprocessing around the abnormality.

Source	Main contribution	Relevance to this project	Limitation / gap
Shen et al. / GMIC	Proposes a global-local architecture that analyses the full mammogram, selects informative local patches, and fuses global and local features for prediction.	Strongly supports the idea that both local lesion details and global mammographic context may contribute to classification. This motivates a hybrid or comparative approach.	GMIC is a complex weakly supervised architecture designed for high-resolution screening mammography. It is not a direct controlled comparison of crop, mask, and contextual ROI representations on CBIS-DDSM mass CADx.

Abdikenov et al. reduce the effective dataset size by pairing CC and MLO views. The CBIS-DDSM dataset primary key is the case, an aggregation of the patient id, the view and the nth occurrence the exam was conducted under these conditions. The study entails a reduction of the training set by consolidating CC and MLO views per patient, per occurrence. Since paired views are not available for every case, it may account for a significant loss of information. CNN training generally requires sufficiently large and representative datasets. Although this study acknowledges the benefit of a multi-angle approach, it is therefore less suitable for the present dataset. Shen et al. offer a solution by introducing a full context / cropped dichotomy. This applies at case level and provides complementary global and local information to the classifier. However, it entails an enhanced preprocessing due to the cropping engineering. Zooming over the lesion raises questions about the margins and the image resolution.

This review identifies two connected research questions. First, how does lesion representation - full mammogram or ROI-centred crop - affect benign-malignant classification under a controlled CNN baseline? Second, once an effective local representation has been selected, does combining it with a lightweight global branch improve performance over either input alone?

2.2.1.2 Model Building a CNN for mammographic classification raises the question of whether features should be learned exclusively from the target dataset or transferred from a model pretrained on a larger image corpus. The reviewed mammography studies commonly use architectures such as ResNet, DenseNet and EfficientNet.

Raghu et al. question the extent to which high-level ImageNet features are reused in medical-image classification. Their experiments suggest that transferable information may be concentrated mainly in the earlier layers and that smaller models trained from scratch can sometimes achieve performance comparable to larger pretrained architectures (Raghu et al. 2019). These findings support the use of a lightweight CNN as a meaningful baseline, rather than demonstrating that training from scratch is inherently superior.

Mammograms may contain annotations or acquisition artefacts that support shortcut learning. Preprocessing controls are thus required to remove visible non-anatomical information, while Grad-CAM visualisations can provide complementary qualitative evidence about the regions influencing a prediction. Raghu et al. do not make any mention of preprocessing to clean the input medical images. Assuming annotations remain, lightweight and transfer-learning models may converge in terms of performance.

Liao and Aagaard evaluated ImageNet-pretrained ResNet50 and Xception models using ROI-centred crops from the CBIS-DDSM mass subset. Their 448×448 -pixel ResNet50 configuration achieved the strongest malignant-case detection among their tested configurations. Their experiments suggested that excessive downsampling could remove relevant lesion information (Liao and Aagaard 2024). This provides direct support for evaluating ResNet50 with a moderately high-resolution local input in the present project.

However, the suitability of ResNet50 remains an empirical question. ImageNet contains natural colour images that differ substantially from grayscale mammograms, and performance on the official CBIS-DDSM split does not demonstrate external generalisation to other institutions or acquisition systems. Islam et al. further demonstrate that ResNet50 can be adapted using task-specific components such as convolutional attention (Islam et al. 2022), although this does not establish its superiority over other pretrained architectures.

The custom CNN and ResNet50 therefore represent two complementary learning strategies. The custom CNN has lower capacity and learns exclusively from CBIS-DDSM, whereas ResNet50 provides a richer pretrained representation at the cost of greater complexity and domain mismatch. Neither approach is assumed to be superior in advance. The retained architecture is selected through controlled validation experiments.

2.2.1.3 Fusion Combining the local and global branches raises a fundamental architectural question: should the final prediction be obtained by combining two independently estimated probabilities, or should it be learned from a joint representation of local and global features?

Decision-level fusion combines the outputs of separately trained classifiers, for example by averaging their predicted probabilities. This approach is simple and preserves the autonomy of each branch, but it provides limited capacity to model interactions between each other. Feature-level fusion instead combines

intermediate representations before applying a shared classification head. It can therefore learn whether particular global characteristics modify the interpretation of specific local lesion features.

Shen et al. examined both strategies in GMIC. Their preliminary architecture averaged the predictions produced by the global and local modules, whereas the extended model introduced a learned fusion module combining a global mammographic representation with an attention-weighted representation of selected local patches (Shen et al. 2021). Their ablation experiments showed that the learned fusion prediction consistently achieved a higher AUC than either branch alone or the average of their probabilities.

The present project differs from GMIC because lesion locations are provided by expert ROI masks rather than inferred from weakly supervised saliency maps. Nevertheless, GMIC supports the principle that global context and fine-grained local information should be integrated before producing the final prediction.

A simple concatenation nevertheless gives both representations nominally equal access to the classification head without ensuring that they contribute equally or usefully. Differences in feature dimensionality, scale and predictive strength may allow one branch to dominate the learned decision. The present project therefore also evaluates an asymmetric residual fusion strategy. The pretrained local classifier supplies the baseline logit, while a lightweight head uses the local and global embeddings to learn a contextual correction. This design treats lesion morphology as the primary source of evidence and the full mammogram as complementary context, rather than requiring both branches to make equivalent independent contributions.

2.2.2 Evaluation

Varoquaux and Cheplygina stress that medical imaging studies require meaningful baselines, because performance claims are difficult to interpret if the model is not compared against an appropriate reference (Varoquaux and Cheplygina 2022). A baseline is therefore needed. Chollet’s deep learning workflow provides a practical framework for implementing a binary classification model. It consists of beginning with a simple working model, monitoring validation performance, diagnosing overfitting or underfitting, and only then increasing model complexity.

Metric choice also affects how model performance is interpreted. Accuracy is frequently used as a primary metric to assess the model’s performance. Abdikenov et al. emphasise this metric when presenting a model. Their study emphasizes that “the accuracy rose to 88.53%, with a remarkably high ROC AUC of 0.9283.” Although it is an interesting metric, false negatives and false positives do not carry the same clinical consequences in the medical profession. Varoquaux and Cheplygina therefore insist that “important metrics may be missing from evaluation. Next to typical classification metrics (sensitivity, specificity, area under the curve), several authors argue for a calibration metric that compares the

predicted and observed probabilities”.

For this reason, the present project does not rely on accuracy alone. ROC-AUC is used for threshold-independent comparison because it measures the model’s ability to rank malignant cases above benign cases. Recall is particularly important because false negative predictions have more consequences than false positive.

Binary cross-entropy is monitored to assess the quality of probabilistic predictions. Unlike accuracy or AUC, it penalises models that assign high confidence to incorrect classifications. However, binary cross-entropy does not provide a complete assessment of calibration. The final evaluation should therefore also examine whether predicted probabilities correspond appropriately to observed outcome frequencies. Interpretability is assessed separately through Grad-CAM visualisations, which provide qualitative evidence about the image regions contributing to the model’s predictions.

3 Design

3.1 Domain and users

In the project scope, the presence of an abnormality is already known and needs to be assessed. The primary users would be radiologists who would use it as a decision-making tool to assess the nature of a tumour.

From this perspective, the user and the target population should be differentiated. Such a model would support specialists by providing a second-opinion signal, helping prioritise uncertain cases, and flagging predictions that require closer review. The pipeline could assist with triage while supporting the review of uncertain cases. The level of human oversight should remain high, particularly for uncertain or borderline predictions. Such pipeline therefore needs to provide transparency and explainability to point towards cases that require further examination. Rezazade Mehrizi et al. emphasize the importance of decision system evaluation. AI suggestions can affect radiologists’ decisions in mammography examination, making explainability tools central to clinical deployment, especially visual heat maps (Rezazade Mehrizi et al. 2023).

Although breast cancer can also occur in men, male cases account for only around 0.5–1% of breast cancer cases. The target population represented by mammography screening datasets is therefore predominantly female.

3.2 Dataset

The model is trained on the CBIS-DDSM dataset. Each abnormality is associated with a cropped image centred on the lesion, a full mammogram, and a binary mask locating the ROI.

Varoquaux and Cheplygina stress the importance of two risks. First, the train,

validation and test sets should be exempted of patient leakage (Varoquaux and Cheplygina 2022). The CBIS-DDSM dataset uses the lesion as a primary key. The same patient can thus be part of two distinct sets, inflating the performance as a result. To address this risk, the overlap based on the patient ID are investigated between the original train and test sets. This will be achieved using pandas and numpy libraries. If no patient leakage is confirmed, the test set is fixed as a CSV file.

Development initially used fixed patient-grouped train and validation partitions derived from the original CBIS-DDSM training set, while the official test set was kept separate, all of which are saved in CSV files. The validation partition represented approximately 20% of the development data and was selected to preserve malignant prevalence while preventing patient overlap. Because performance estimated from one partition may be unstable on a limited dataset, the final architecture comparison additionally used five-fold patient-grouped cross-validation over the combined training and validation data.

Second, Varoquaux and Cheplygina considers the risk of mislabelled dataset. Because CBIS-DDSM is organised at lesion level, several observations may belong to the same patient. For the global branch, duplicate lesion records belonging to the same mammogram were consolidated using patient ID, laterality and view. When several lesions occurred in one mammogram, the maximum lesion label was retained so that any mammogram containing a malignant lesion was classified as malignant.

3.3 Data pre-processing

3.3.1 Image orientation

Normalizing data input improves the model capability to learn a signal from data. The DICOMs are natively displayed in different positions. Some breasts are rotated to the left, other ones to the right. I decided to orient the breasts to the right side. A black pixel is worth 0. Its score increases to 1 as it tends to a white colour. The breast side will be inferred from the spatial distribution of foreground tissue. First, a binary tissue mask will be obtained by thresholding the mammogram.

$$T(y, x) = \begin{cases} 1, & \text{if } I(y, x) > \tau, \\ 0, & \text{otherwise,} \end{cases}$$

where $I(y, x)$ is the image intensity at position (y, x) and τ is the selected threshold.

The amount of tissue in the left and right halves of an image of width w will then be computed as:

$$M_{\text{left}} = \sum_y \sum_{x=0}^{\lfloor w/2 \rfloor - 1} T(y, x),$$

$$M_{\text{right}} = \sum_y \sum_{x=\lfloor w/2 \rfloor}^{w-1} T(y, x).$$

The breast will be considered to lie on the left-hand side of the image when:

$$M_{\text{left}} > M_{\text{right}}.$$

Otherwise, it will be considered to lie on the right-hand side.

The process was implemented using numpy library.

3.3.2 Cropping

The provided cropped image was not used because it offers limited control over the margin surrounding the lesion. To improve the normalization across cropped image, I will use the provided binary mask to derive the ROI coordinates. This project is restricted to the CADx task. Cropping using masking coordinates is a non-greedy algorithm that fits this scope.

Liao et al. state that cropping entails a trade-off between resolution and generalization (Liao and Aagaard 2024). I will apply a generic margin by calculating the geometric center of the lesion and standardizing the margins from there accordingly. This study standardises crop dimensions to improve the consistency and generalisability of the input representation. Defining the cropped area is done with numpy and the result is stored as a tensor using TensorFlow.

The geometric center of the lesion is calculated as follows:

$$x_c = \frac{\sum_{x=1}^W \sum_{y=1}^H x M(x, y)}{\sum_{x=1}^W \sum_{y=1}^H M(x, y)}, \quad y_c = \frac{\sum_{x=1}^W \sum_{y=1}^H y M(x, y)}{\sum_{x=1}^W \sum_{y=1}^H M(x, y)} \quad (1)$$

The resulting centre coordinates are represented as:

$$C_{\text{ROI}} = (x_c, y_c) \quad (2)$$

The coordinates are then applied to the full mammogram. The resulting area is resized to a 384x384-pixel image.

3.3.3 Full mammogram curation

Annotations are sometimes handwritten on the non-breast area. There is a risk the model learns these patterns and develop biases as a result when classifying new inputs. Such artefacts are removed using OpenCV. Thresholding separates foreground structures from the background, while morphological operations reconnect breast tissue and remove small isolated patterns. Connected-component analysis then retains the most plausible breast region. Finally, the image is cropped according to the target aspect ratio and resized to the model input dimensions, while limiting geometric distortion and tissue loss.

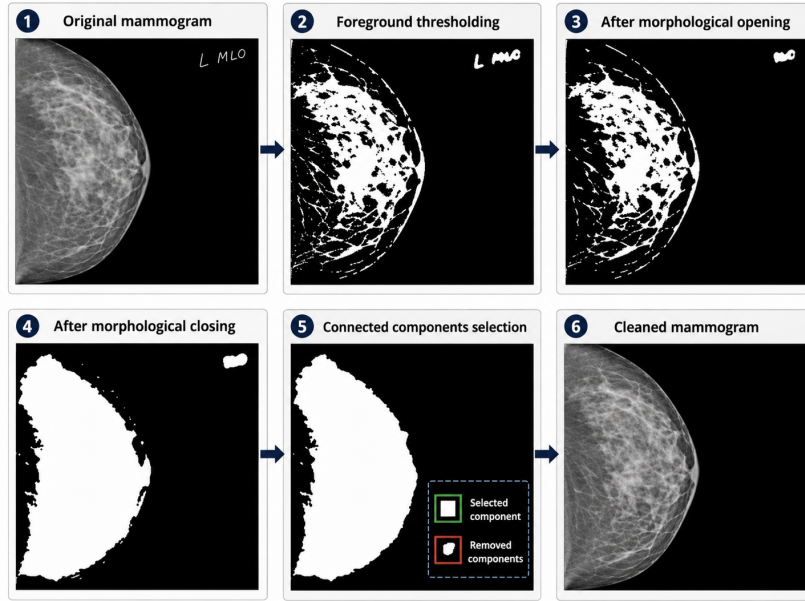


Figure 1: Annotation removal example

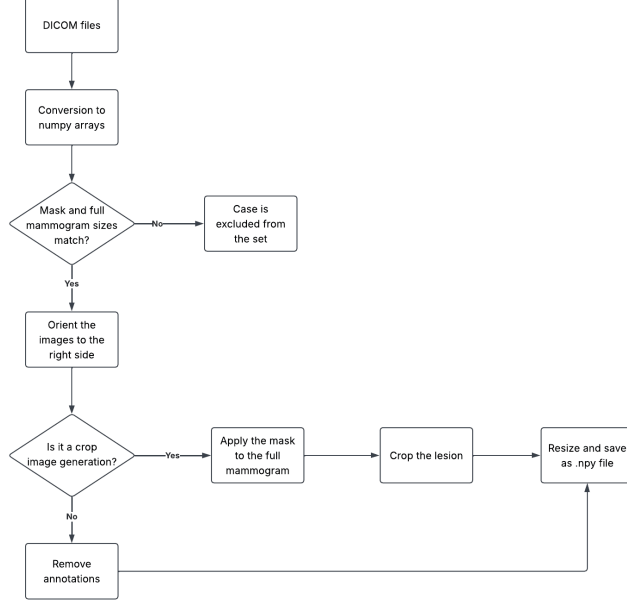


Figure 2: Pipeline

3.4 Architecture

3.4.1 Local and global branches

Two modelling strategies were compared for both branches, a lightweight CNN trained from scratch on CBIS-DDSM and an ImageNet-pretrained ResNet50. The lightweight CNN provided a controlled low-capacity baseline, while ResNet50 provided a richer pretrained representation at the cost of greater complexity and domain mismatch. Input dimensions were fixed at 384×384 pixels for the local crop and 768×512 pixels for the full mammogram. Model capacity was increased progressively when indicated by validation performance.

Raghu et al. emphasized the gain from transfer-learning models was not significant compared to lightweight ones. A first architecture is attempted for both local and global branches. The baseline architecture begins with a single convolutional block. The output activation function is sigmoid, because the classification is binary.

$$y = \frac{1}{1 + \exp(-x)}$$

Due to the lack of available data in CBIS-DDSM, this baseline may reach a performance ceiling, even after optimization. Improving it entails the use of trans-

fer learning techniques. ResNet50 is evaluated as the transfer-learning backbone because it provides a computationally manageable residual architecture.

3.4.2 Fusion

The fusion design examines two alternative relationships between the local and global branches.

The first strategy uses symmetric feature fusion. The intermediate feature vectors produced by the frozen local and global branches are normalised and concatenated. A shared classification head then learns a malignancy probability from the joint representation. This approach allows interactions between local morphology and global mammographic context, but it does not guarantee that the two branches contribute equally or usefully.

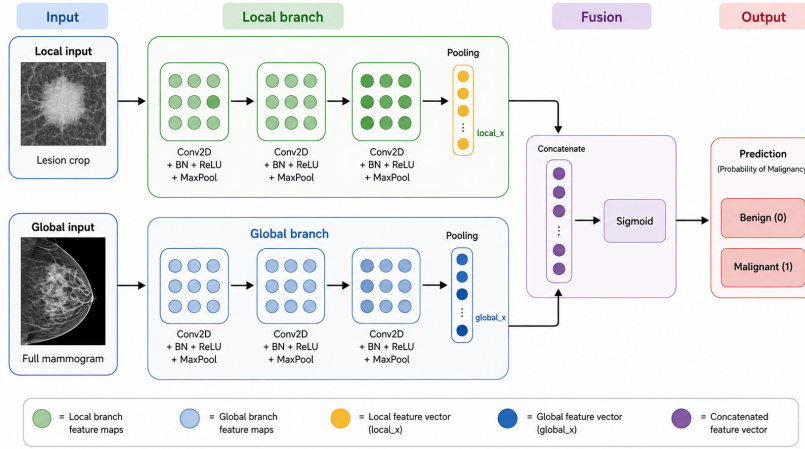


Figure 3: Fusion model architecture

The second strategy uses asymmetric residual fusion. The local classifier provides the baseline prediction because its input is centred directly on the annotated lesion. The local and global embeddings are concatenated and passed through a lightweight correction head, which produces a contextual correction in logit space. This correction is then added to the frozen local pre-sigmoid logit:

$$z_{\text{final}} = z_{\text{local}} + \Delta z_{\text{context}}$$

where z_{local} is the local-branch logit and $\Delta z_{\text{context}}$ is the correction learned from the combined local and global embeddings. The final probability is obtained by applying the sigmoid function to z_{final} .

The correction output is initialised to zero, so the initial fusion prediction is identical to the pretrained local prediction. Training therefore evaluates whether global context can improve the local decision without requiring the fusion model to relearn the existing local classifier from scratch.

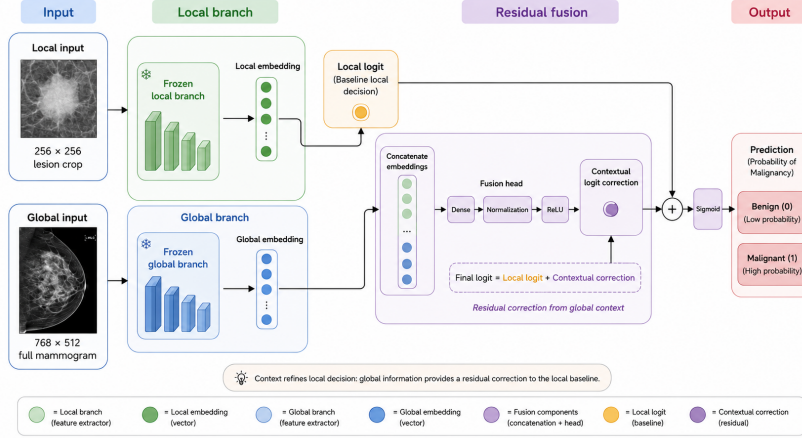


Figure 4: Context logit model architecture

3.5 Work plan

The project will be developed in four main phases: data preparation, representation selection, final modelling, and evaluation/reporting.

Task	June	July	August	September
Dataset ingestion and metadata matching	████			
ROI-mask alignment and visual checks	████			
Preprocessing comparison	████	████		
Select best image representation		████		
Train improved single-branch CNN		████	████	
Implement simplified global-local model		████	████	
Tune regularisation and augmentation			████	
Evaluation and interpretability outputs			████	████
Final report and figures			████	████

Table 2: Planned project timeline

3.6 Testing and evaluation

3.6.1 Generic model strategy

Model evaluation considers both discrimination and the quality of probabilistic predictions. ROC-AUC is used as the primary threshold-independent discrimi-

nation metric because it measures the model’s ability to rank malignant lesions above benign lesions. An ROC-AUC of 0.5 corresponds to chance-level ranking.

Binary cross-entropy is monitored because it evaluates the probability assigned to the observed class and penalises confident incorrect predictions. It is compared with the null loss obtained from a constant classifier that predicts the malignant-class prevalence for every sample:

$$\text{BCE}_{\text{null}} = -[\pi \log(\pi) + (1 - \pi) \log(1 - \pi)]$$

For the validation set, the malignant-class prevalence is approximately 0.48, corresponding to a null BCE of approximately 0.693. A validation BCE below this value indicates that the model provides more informative probabilistic predictions than the prevalence-only baseline.

Training and validation ROC-AUC and binary cross-entropy curves are inspected jointly to identify underfitting and overfitting. The curves are generated with matplotlib library. The retained checkpoint is selected using validation loss. Accuracy, sensitivity, specificity and precision are evaluated after selecting a decision threshold on the validation set. Threshold selection prioritises sensitivity while maintaining acceptable precision.

3.6.2 Fusion strategy

The value of fusion is assessed against the strongest individual branch rather than only against the initial lightweight CNN.

During model training, binary cross-entropy (BCE) is used as the optimisation objective and validation loss is used for checkpoint selection. Consequently, the retained checkpoint for each model is the one achieving the lowest validation BCE. ROC-AUC is not used for checkpoint selection.

The two fusion strategies are compared using five-fold patient-grouped cross-validation on the development data (train and validation sets added). The same outer patient partitions and the same training seed are used for both architectures. This design evaluates whether differences between the symmetric and residual fusion strategies are consistent across different held-out patient groups.

ROC-AUC is used to characterise threshold-independent discrimination, while BCE assesses probabilistic predictive performance and strongly penalises confident incorrect predictions. Average precision (AP) is retained as a complementary discrimination measure with greater emphasis on performance for the malignant class. The Brier score is additionally reported as a complementary quadratic measure of probabilistic accuracy. Its bounded quadratic penalty is less strongly influenced by individual highly confident errors than BCE and permits an interpretable comparison with a prevalence-only probabilistic predictor.

Because neither BCE nor the Brier score constitutes a complete assessment of calibration, calibration is evaluated separately using out-of-fold predictions.

Since both fusion architectures are evaluated on the same outer-fold observations, their performances are also compared using paired metric differences. For a metric M , the difference is defined as

$$\Delta M = M_{\text{residual}} - M_{\text{symmetric}}$$

Positive values of delta ROC-AUC and delta AP favour residual fusion, whereas negative values of delta BCE and delta Brier favour residual fusion. After completion of all outer folds, out-of-fold predictions are used to estimate uncertainty in these paired differences at patient level.

The same evaluation metrics will be applied to the local and global branches on matched outer-fold observations. This will allow the fusion models to be compared not only with each other, but also with the strongest single-branch baseline.

4 Implementation

4.1 Project tree

```
src/
|-- data/
|   |-- build_dataset.py
|   |-- split_evaluation.py
|   |-- train_val_test_split.py
|   |-- unique_patient_global.py
|-- preprocessing/
|   |-- dataset_preprocessing.py
|   |-- dicom_handling.py
|-- modeling/
|   |-- fusion.py
|   |-- global_resnet50.py
|   |-- local_resnet50.py
|-- training/
|   |-- cnn_evaluation.py
|   |-- dataset_preparation.py
|   |-- threshold_selection.py
|   |-- resnet50_global_branch.py
|   |-- resnet50_local_branch.py
|   |-- fusion_global_local_resnet50.py
|-- predict/
|   |-- evaluate_residual_threshold.py
|   |-- evaluate_saved_model.py
|-- evaluation/
```

```

| |-- analyse_oof.py
| |-- evaluation_metrics.py
| |-- generate_branch_oof.py
| `-- run_fusion_cv.py
|-- xAI_gradcam/
| |-- grad_cam_residual_resnet.py
| `-- grad_cam_fusion.py
|-- config.py
`-- functions.py

```

4.2 Inputs

4.2.1 Dataset creation

The CBIS-DDSM original package is made of DICOMs representing the mammograms and two CSV files - `mass_case_description_train_set.csv` and `mass_case_description_test_set.csv` - pointing to their location for every abnormality case.

The file names disclosed in the CSV do not match the actual one even though the given paths point to the correct folders. Building the dataset first requires to resolve the file names. For every case, for the fields “ROI mask file path”, “image file path” and “cropped image file path” the files located in these paths are retrieved and their nature is assessed using `pydicom.dcmread`, which allows to retrieve the parameter `SeriesDescription`, stating the nature of the file. Once resolved, the actual paths are added to the dataset. Two fields are also added. “Label” converts the severity of the lesion into a binary. “BENIGN” and “BENIGN_WITHOUT_CALLBACK” are worth 0 and “MALIGNANT”, 1. If the label returns N/A, the row is excluded (see Appendix 1).

Due to the risk of data leakage, The lesion-level partitions are stored as `train_split.csv`, `val_split.csv` and `test_split.csv`. Equivalent mammogram-level partitions are generated for the global branch after consolidating duplicate lesion records belonging to the same mammogram. These fixed partitions ensure the experiments are conducted under comparable conditions and can be reproduced. The train / validation split is made based on a traditional 80% / 20% ratio of the train set. Both sets should contain an equivalent proportion of malignant lesions. Several splits are tested and the candidate with the lowest score is retained. The score maximizes the equivalence in terms of malignancy proportion while sticking to the 80/20 ratio (see Appendix 2).

```

StratifiedGroupKFold(
    stratify = pathology,
    group = patient_id
)

```

```

# Select the candidate split minimizing:
# |p_train - p_all| + |p_val - p_all| + size_error

```


There is a check conducted ex-post to confirm there is no patient overlap.

```
train_patients = set(train_df["patient_id"].dropna())
val_patients = set(val_df["patient_id"].dropna())
train_val_overlap = (train_patients & val_patients)
```

Because the training and validation sets are approximately balanced between benign and malignant cases, class weighting was not applied.

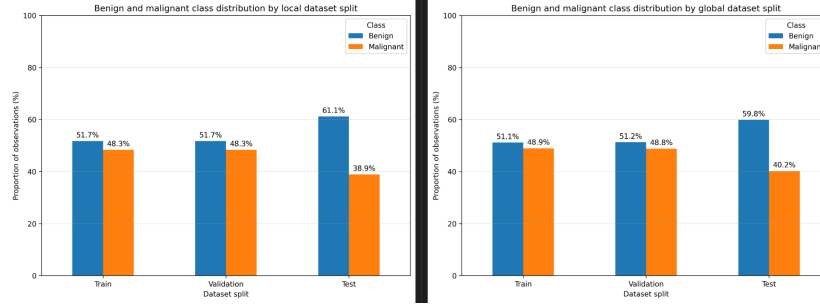


Figure 5: Global and Local label distribution across splits

4.2.2 Preprocessing

4.2.2.1 Shared preprocessing DICOM pixel arrays were preprocessed once and stored as NumPy arrays to avoid repeated conversion during model training using pydicom library. Images were normalised to a common right-facing orientation before branch-specific preprocessing to normalize the reading. It is achieved by converting the 2D array into a boolean one that returns True for every pixel exceeding a threshold value that defines whether the location may represent tissues. The side with tissue is pivoted to the right (see Appendix 3).

4.2.2.2 Local inputs The mask is used to locate the tumour in the array. As a prerequisite, the mask and the image sizes are confirmed to be the same. If not, the case is excluded from the set, logged in a CSV file named “logs_history.csv” and a log is sent to the terminal for review. The total number of excluded cases is returned after the files were created.

Mask files that pass the check have the geometric center of the lesion calculated. A square 384×384-pixel crop is centred on the lesion centroid. The array needs to have a third dimension added to be used by the CNN. It is added using `expand_dims`, from the numpy library, and converted into tensor with tensorflow. The object is saved as .npy file under a newly created or reset folder.

When successful, the processed zoom image is given by the image on the upper-right corner.

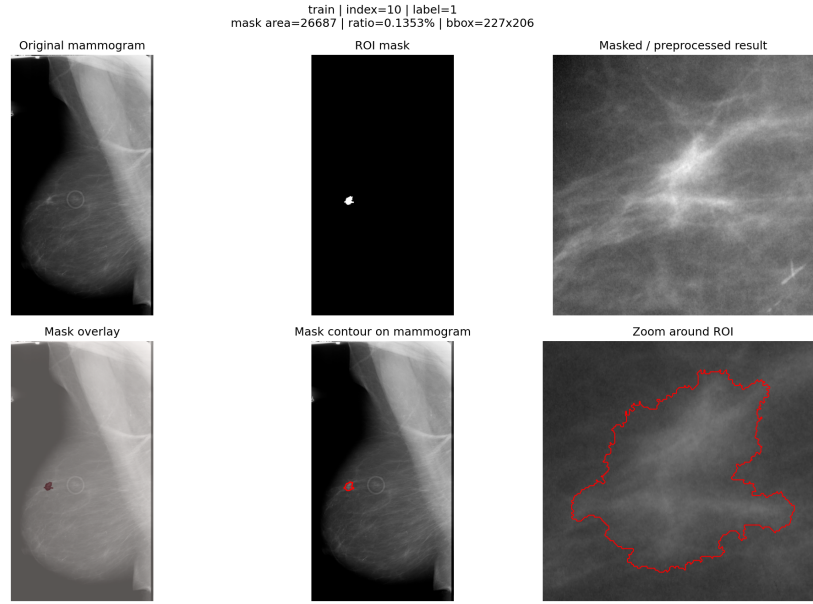


Figure 6: Zoom image

4.2.2.3 Global inputs Global input arrays are also subject to preprocessing, because annotations need to be removed. I used an OpenCV-based method combining thresholding, morphological operations and connected-component analysis. This refined approach aims to retain the main breast region while removing smaller isolated annotations and background artefacts.



Figure 7: Image curated from its annotations

4.2.2.4 Numpy file dataset A CSV file is also created that resolves the paths of these newly created numpy files. There is one CSV per preprocessing type because they point to different folders depending on their preprocessing

specifics.

4.3 Training pipeline and reproducibility

4.3.1 Local branch parameters

4.3.1.1 Baseline CNN The local branch receives a 384×384 -pixel crop centred on the lesion. The first convolutional layer contains 32 learnable filters, producing 32 feature maps representing low-level local patterns.

Table 3: First local-branch experiment.

Parameter	Initial value
Input	ROI-centred lesion crop
Input shape	$(384 \times 384 \times 1)$
Batch size	16
Maximum epochs	100
Convolutional blocks	1
Convolutional filters	32
Convolutional activation	ReLU
Feature aggregation	Flatten
Output activation	Sigmoid
Loss function	Binary cross-entropy
Model selection metric	Validation loss
Early stopping patience	5 epochs

To prevent overfitting, the minimum validation loss is used for early stopping after six epochs. The process consists in increasing the model capacity until it shows signs of overfitting. Eventually, the model is run on three convolutional blocks 32, 64 and 128 (see Appendix 4).

Flatten was evaluated because it preserves all spatial feature activations rather than reducing each channel to a single summary value. However, this produces a high-dimensional representation and therefore increases the risk of overfitting compared with global pooling.

The training uses a validation set and stops five epochs after it reached its lowest loss. The best epoch is saved in a checkpoint keras file, along with the AUC and loss curves of the train and validation sets. Yet, the model did not achieve a validation BCE below the prevalence-based null baseline of approximately 0.693.

4.3.1.2 Transfer-learning models I tested a naive ResNet50 implementation. ResNet50 accepts 3-channel inputs. The same tensor was passed three times to comply (details in Appendix 5).

The ResNet50 backbone was frozen. Its convolutional weights were therefore not updated from CBIS-DDSM, while the newly added classification head learned a task-specific mapping from the pretrained feature representation.

Data augmentation was introduced to increase training variability and reduce sensitivity to nuisance transformations. I used horizontal flipping as a data-augmentation operation. This increased the effective diversity of the training data and encouraged the classifier to reduce its sensitivity to the horizontal orientation of the lesion crop.

I considered two additional features. First, MaxPooling2D was added. The additional layer contributes to reduce the dimensionality of the representation while preserving the strongest local activation within each pooling window and feature channel.

Second, a Dense layer containing 16 ReLU-activated units was placed between the flattened ResNet representation and the dropout layer. This layer acted as a trainable task-specific adapter. It compressed the pretrained feature representation and introduced a nonlinear transformation before the final sigmoid classifier (see Appendix 6).

The baseline CNN, frozen ResNet50 configuration, augmentation pipeline, spatial pooling operation and task-specific Dense adapter constituted successive implemented variants. Their effects on validation loss, AUC, generalisation and overfitting are compared in the evaluation chapter.

The retained architecture for the local branch is as per the below.

Parameter	Initial value
Input	ROI-centred lesion crop
Input shape	$(384 \times 384 \times 1)$
Batch size	16
Maximum epochs	100
Data augmentation	Random horizontal flip
Input conversion	Grayscale channel replicated into 3 channels
Input rescaling	Pixel values multiplied by 255
Backbone model	ResNet50
Pretrained weights	ImageNet
Classification top	Excluded (<code>include_top=False</code>)
Backbone training	Frozen
Spatial feature reduction	(4×4) max pooling
Pooling stride	(4×4)
Pooling padding	Same
Feature aggregation	Flatten
Adapter layer	Dense layer with 16 units
Adapter activation	ReLU
Adapter regularization	L2, (1×10^{-5})

Parameter	Initial value
Classification dropout	0.5
Output units	1
Output activation	Sigmoid
Output regularization	L2, (1×10^{-5})
Loss function	Binary cross-entropy
Optimizer	Adam
Initial learning rate	(1×10^{-4})
Learning-rate scheduler	ReduceLROnPlateau
Learning-rate reduction	Factor 0.2 after 3 epochs without improvement
Minimum learning rate	(1×10^{-6})
Model selection metric	Validation loss
Model selection mode	Minimum
Early stopping patience	6 epochs
Best-model checkpoint	Lowest validation loss

4.3.2 Global branch parameters

4.3.2.1 Baseline CNN Mirroring the approach for the local branch, the first experiment for the global branch was conducted using the configuration shown below.

Parameter	Initial value
Input	Full mammogram
Input shape	$(768 \times 512 \times 1)$
Batch size	16
Maximum epochs	100
Convolutional blocks	1
Convolutional filters	16
Convolutional activation	ReLU
Feature aggregation	Flatten
Output activation	Sigmoid
Loss function	Binary cross-entropy
Model selection metric	Validation loss
Early stopping patience	6 epochs

This architecture starts with one 16-filter convolutional block. The model is expected to learn broad-based patterns from the full image that can complement those learnt by the local branch. Flatten was retained for feature aggregation because it preserves the spatial feature representation. The python implementation is strictly similar to the local branch's.

4.3.2.2 Transfer-learning model Several ResNet50 architectures were tested. The first versions were very similar to those described above, for the

local branch.

Given the larger spatial dimensions of the global input, a 6×6 pooling operation was evaluated to reduce the dimensionality of the flattened representation and limit model capacity.

A Dense-BatchNormalization-ReLU block was introduced as a trainable task-specific adapter. The Dense layer compressed the flattened representation, while batch normalization stabilised the activation distribution before applying the non-linearity. This layer is intended as a trainable task-specific adapter (details in Appendix 7).

Several data augmentation strategies were also tested on the branch, such as RandomTranslation or RandomFlip. RandomRotation was ruled out due to the effort put to normalize the image and rotate the breast on the right.

Other architectures were tested that adjusted the dropout rates and the pool_size.

The retained architecture for the global branch is as per the below.

Parameter	Initial value
Input	Full-breast mammogram
Input shape	$(768 \times 512 \times 1)$
Batch size	16
Maximum epochs	100
Data augmentation	None
Input conversion	Grayscale channel replicated into 3 channels
Input rescaling	Pixel values multiplied by 255
Backbone model	ResNet50
Pretrained weights	ImageNet
Classification top	Excluded (<code>include_top=False</code>)
Backbone training	Frozen
Spatial feature reduction	(6×6) max pooling
Pooling stride	(6×6)
Pooling padding	Same
Feature aggregation	Flatten
Classification dropout	0.5
Adapter layer	Dense layer with 8 units
Adapter bias	Disabled
Adapter regularization	L2, (1×10^{-4})
Adapter normalization	Batch normalization
Adapter activation	ReLU
Embedding dropout	0.2
Output units	1
Output activation	Sigmoid
Loss function	Binary cross-entropy

Parameter	Initial value
Optimizer	Adam
Initial learning rate	(1×10^{-4})
Learning-rate scheduler	ReduceLROnPlateau
Learning-rate reduction	Factor 0.2 after 3 epochs without improvement
Minimum learning rate	(1×10^{-6})
Model selection metric	Validation loss
Model selection mode	Minimum
Early stopping patience	6 epochs
Best-model checkpoint	Lowest validation loss

4.3.3 Fusion model

A ResNet50 fusion required to merge embedding features. In this case, the retrieved layers are Dense16 for local branch and Dense8 for the global one. The dimensions are (None, D). Proceeding as such creates a unique lesion / probability pair, which is achieved by retrieving the full mammogram representing the lesion. In case the global context is associated with different labels, the malignant assessment prevails. The architecture aims at synergizing the local and global features, assuming they carry a different set of information (details in Appendix 8).

Another architecture was tested that treated the local module as the main branch and used the global branch to give additional context, to refine the decision. The root function is similar to the ones described above, the embedding features are still merged. However, the fusion is converted to a logit to have it compared to the local branch decision, which prevails. The result is then converted to a probability. The following pseudo-code summarizes this approach detailed in appendix 9.

```

features = concatenate([local_embedding, global_embedding])
correction = correction_network(features)

contextual_logit = Dense(
    1,
    kernel_initializer="zeros",
    bias_initializer="zeros",
)(correction)

final_logit = local_logit + contextual_logit

```

4.4 Final experimental pipeline

4.4.1 Fusion model selection

The evaluation pipeline used a two-level patient-grouped cross-validation procedure to preserve patient independence and estimate generalisation without using the official test set for architecture selection. The outer folds were used to estimate model generalisation. In each outer iteration, one group of patients was held out entirely for evaluation, while the remaining patients formed the outer training set. The outer evaluation patients were not used for model fitting or checkpoint selection. Within each outer training set, an additional inner split was created to separate the available training patients into an inner training subset and an inner validation subset. The following pseudo-code illustrates the logic detailed in appendix 10.

```
For each outer patient-grouped fold:
    outer_eval ← held-out patients
    outer_train ← remaining patients

    split outer_train by patient into:
        inner_train
        inner_validation

    train on inner_train
    early-stop on inner_validation
    evaluate once on outer_eval
```

```
Patients
|
|--- Outer train
|   +--- Inner train : fitting
|   `--- Inner val   : early stopping
|
|--- Outer eval : OOF prediction
```

The inner training subset was used to fit the model, while the inner validation subset was used for training-time decisions such as early stopping with 6-epoch patience and checkpoint selection based on minimum validation BCE.

Predictions from the symmetric and residual fusion models were then stored for the same held-out observations, enabling direct paired comparison and subsequent patient-level bootstrap analysis. For each fold, the AUC, BCE, average precision and the Brier score are calculated.

4.4.2 Decision Threshold

The threshold selection is operated through a grid search using this method expressed in the following equation.

$$t^* = \arg \max_t \text{Specificity}(t) \quad \text{subject to} \quad \text{Sensitivity}(t) \geq 0.85$$

Because medical classification involves a trade-off between sensitivity and specificity, the decision threshold is selected on the validation set with particular attention to maintaining high sensitivity while preserving a high recall. In this case, the minimum recall is defined as 85%. The technical implementation is available in appendix 11.

4.4.3 Interpretability

Grad-CAM was implemented directly on the residual fusion model by extracting the final convolutional feature maps of both nested ResNet50 branches and reconstructing the forward path to the final residual logit. GradientTape computed class-specific gradients for the local and global feature maps, producing one heatmap per branch. The reconstructed prediction was verified against the loaded model output within 10^{−5}. Global heatmaps were additionally compared with the expert ROI mask using localisation and overlap metrics (see Appendix 12).

```
Local ResNet feature maps ----\
                                +--> Residual model --> Final logit
Global ResNet feature maps ----/
```

5 Evaluation

5.1 Evaluation Overview

The chosen model has its threshold calculated based on the validation set. It is then applied on the test set. Depending on the threshold, the precision, the recall, the accuracy and the F1-score are calculated based on these formula.

Let:

- (TP) denote true positives;
- (TN) denote true negatives;
- (FP) denote false positives;
- (FN) denote false negatives.

5.1.1 Accuracy

Accuracy represents the proportion of correctly classified observations among all observations:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

5.1.2 Precision

Precision represents the proportion of observations predicted as positive that are actually positive:

$$\text{Precision} = \frac{TP}{TP + FP}$$

5.1.3 Recall

Recall, also referred to as sensitivity, represents the proportion of actual positive observations that are correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

5.1.4 F1-score

The F1-score is the harmonic mean of precision and recall:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It can also be expressed directly from the confusion matrix values:

$$F_1 = \frac{2TP}{2TP + FP + FN}$$

5.2 Single-branch evaluation

5.2.1 Local branch

5.2.1.1 Results summary

Stage	Main configuration	Validation			Main finding
		Mean Seeds	AUC	BCE / loss	
Custom CNN base-line	384×384, 1 convolutional block, GlobalAveragePooling	42, 123	0.620	Above / close to null baseline	Predictions were strongly compressed around 0.5, indicating weak discrimination.

Stage	Main configuration	Validation			Main finding
		Seeds	Mean AUC	BCE / loss	
Deeper	384×384, 3 convolutional blocks, MaxPooling, Flatten	42, 123, 999	0.617	Validation loss substantially above training loss	Increasing model capacity did not improve AUC and introduced overfitting.
Frozen ResNet50	GlobalMaxPooling, dropout 0.5	42, 123, 999	0.770	Improved over custom CNN, but still imperfect	Transfer learning produced the first substantial improvement in discrimination.
Fine-tuned ResNet50	Final convolutional block partially fine-tuned	42, 123	0.763	No clear improvement	Fine-tuning did not justify the additional complexity and was discarded.
EfficientNet-B4	ResNet50 + backbone, Flatten, dropout 0.5	42	0.786	Very high	Strong AUC, but extremely confident predictions and poor probabilistic loss.
ResNet50 + Flatten	Frozen backbone, Flatten	42	0.772	Very high	Flatten preserved discrimination but generated highly polarised probabilities and poor BCE.
ResNet50 + 2×2 Max-Pooling	ResNet50, MaxPooling, Flatten	42	0.769	1.3	Spatial pooling reduced the loss by approximately a factor of three while maintaining high AUC, although BCE remained too high.
Lower learning rate	MaxPooling 2×2, Flatten, LR = 1e-4	42, 123	0.764	0.61	Lowering the learning rate largely corrected the loss problem, although overfitting remained visible.

Stage	Main configuration	Validation			Main finding
		Mean SeedsAUC	BCE / loss		
Reduced spatial representation	MaxPooling 4×4, Flatten, L2 regularisation	42, 123, 999	0.782 0.60 0.73	–	Reducing the number of flattened features preserved discrimination while limiting model capacity.
Final local model	ResNet50, MaxPooling 4×4, Flatten, Dense(16), dropout 0.5, L2, RandomFlip	42, 123, 999	0.788 0.58 0.61	–	Best overall trade-off between discrimination and probabilistic loss; relatively stable across seeds, although overfitting emerged after approximately 12 epochs.

5.2.1.2 Key findings

5.2.1.2.1 Can a lightweight CNN learn useful lesion features?

Lightweight CNNs failed to produce sufficiently discriminative validation predictions, and increasing depth introduced overfitting without improving AUC.

5.2.1.2.2 Does transfer learning improve discrimination? To mitigate this lack of signal received from data using lightweight models, I applied a frozen ResNet50 architecture instead. The validation AUC immediately improved to an average 0.770 (+0.160 compared to lightweight architectures). The pre-trained features therefore increased the model’s capability to order cases. The first ResNet50 model relied on a GlobalMaxPooling2D, for performance reasons. GlobalMaxPooling2D has a cost lower than Flatten and helped validating whether ResNet50’s features were relevant in this case.

Although the AUC was improved, the loss did not. A standard ResNet50 model does not manage to improve the baseline BCE. Across the three-seed testing, the loss was consistently around 0.7, indicating that the improved ranking ability did not translate into equally reliable probability estimates.

5.2.1.2.3 Does fine-tuning improve the pretrained representation?

ResNet50 was pretrained on ImageNet, a large natural-image dataset whose visual domain differs substantially from grayscale mammography. ImageNet does not contain images that are directly representative of mammographic lesions.

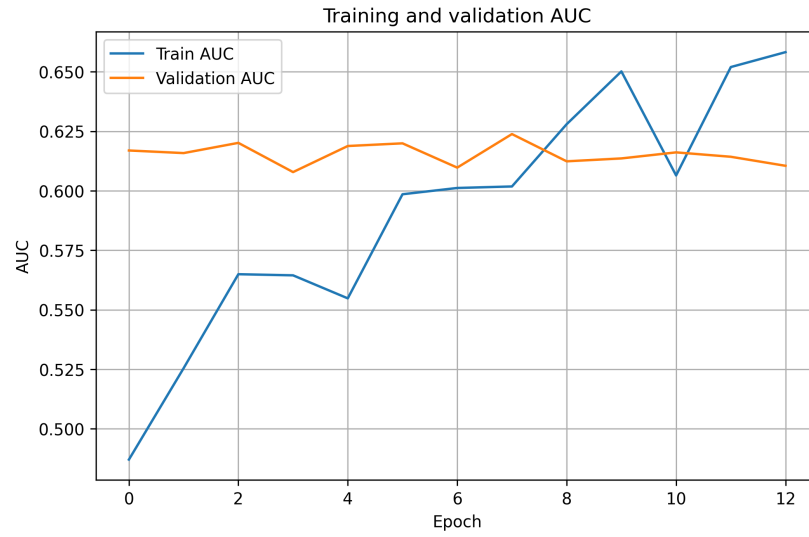


Figure 8: 3-convolutional model AUC

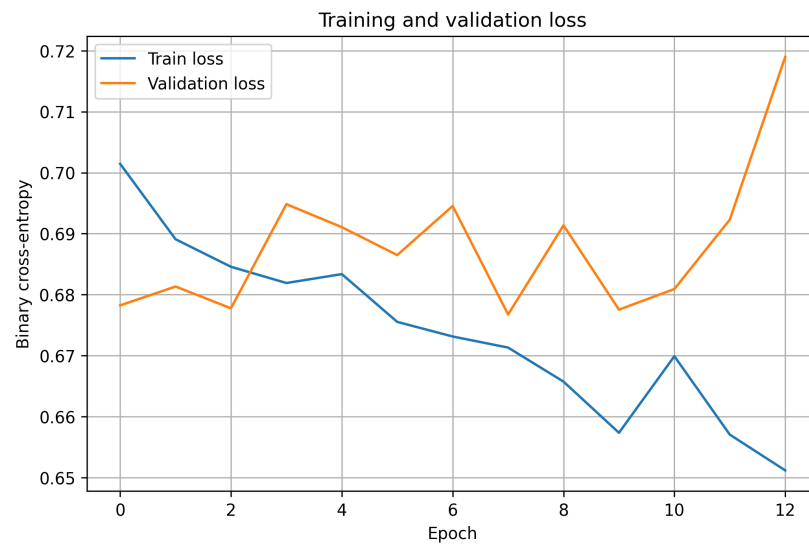


Figure 9: 3-convolutional model loss

Fine-tuning was therefore evaluated to determine whether adapting the higher-level pretrained features to the target domain could improve performance.

Fine-tuning did not provide a consistent validation improvement. Mean validation AUC decreased from approximately 0.770 to 0.763 while validation BCE increased, and validation behaviour was less stable. Fine-tuning was therefore discarded.

5.2.1.2.4 How should pretrained spatial features be aggregated?

GlobalMaxPooling2D did not achieve a validation BCE below the prevalence-based null baseline in the tested configurations. This motivated the evaluation of Flatten. Although it did not degrade the AUC, Flatten worsened the BCE. The loss reached 3.5 on the validation set, almost six times the baseline.

However, in line with the Design choice, Flatten was kept to preserve spatial information. On the other hand, GlobalMaxPooling2D collapses each feature map into a single activation.

Improving the BCE/AUC trade-off was achieved by adding MaxPooling2D before Flatten, with pool size of 2. The architecture reduced the dimensionality of this representation while retaining a coarser spatial organisation. The AUC was preserved to a high score, around 0.770 calculated on the validation set. The BCE decreased to 1.5 on the validation set.

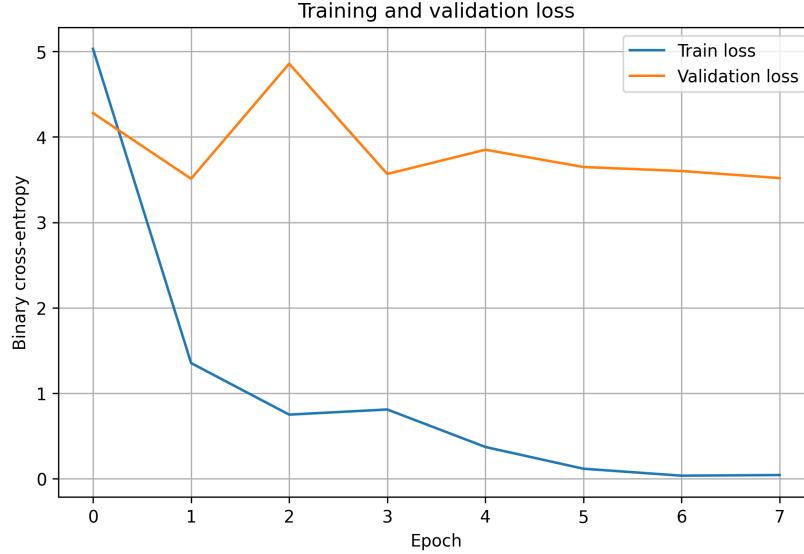


Figure 10: BCE evolution with Flatten

5.2.1.2.5 Can regularisation improve the AUC–BCE trade-off? The previous experiments were carried with a learning rate of 1e-3. Despite the

introduction of ReduceLROnPlateau, the learning rate may not decrease fast enough to learn granularity in patterns.

I reduced the learning rate to $1e-4$ to validate the above assumption. The mean validation AUC remained at 0.764, while validation BCE decreased to approximately 0.61 (-0.08 compared to the baseline). This was the first tested configuration to simultaneously maintain AUC clearly above chance and achieve BCE below the prevalence-based null baseline.

5.2.1.2.6 Does a task-specific trainable adapter improve the pre-trained representation? The previous feature aggregation strategy maintained useful discrimination but left room for improvement in probabilistic error. A Dense16 layer was therefore introduced as a small task-specific adapter between the frozen representation and the classifier.

With an average 0.788 (+0.006 compared to previous best model), the validation AUC remains high and is stable across seeds with this architecture. In addition, the validation BCE decreased to an average 0.58 (-0.11 compared to the baseline). A dense layer improves the ResNet50 model.

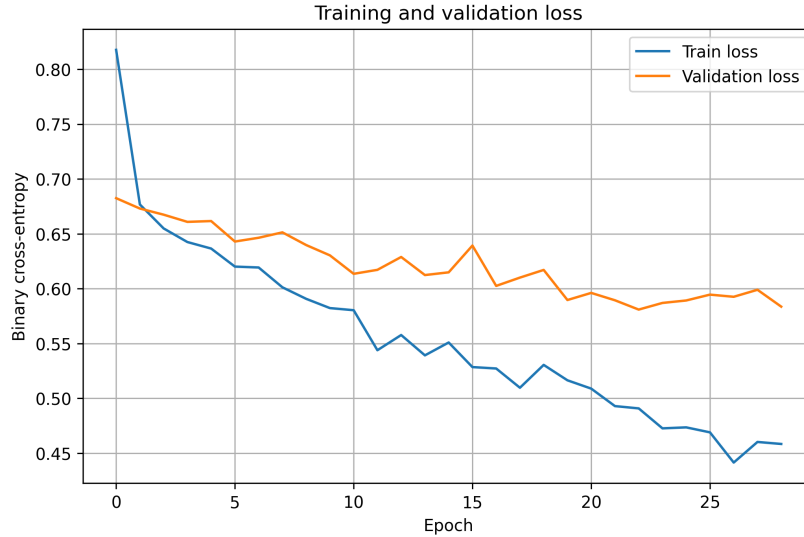


Figure 11: BCE evolution with Dense16

5.2.2 Global Branch

5.2.2.1 Results summary

Stage	Main configuration	Seeds	Mean val. AUC	AUC range	Val. BCE	Main finding
Lightweight base-line	78×512, Conv(16), MaxPool, Flatten	42, 123	0.522	0.515–0.529	1.25	Very weak discrimination; increasing CNN depth did not provide a consistent improvement.
Higher-capacity CNN	Up to 3 conv layers; Dense head tested	42, 123	0.553	—	0.80	Additional capacity improved BCE but discrimination remained close to chance.
ResNet50 base-line	Frozen ResNet50, MaxPool 6×6, Flatten	123	0.617	—	0.90	Transfer learning revealed substantially more predictive signal than the lightweight CNN.
+ Dropout	+ dropout 0.5	123	0.679	—	0.90	Discrimination improved substantially, although BCE remained high.
Naive Dense8 adapter	+ Dense8	42, 123	0.574	0.491–0.656	0.68	BCE improved, but performance was highly seed-dependent.
Structured adapter	Dense8 → BatchNorm → ReLU	42, 123	0.683	0.681–0.685	0.63	Batch normalization stabilised the adapter and produced consistent discrimination.
+ Additional dropout	+ dropout 0.2 before classifier	123	0.693	—	0.63	Small AUC improvement without deterioration in BCE.
Final global model	+ L2 1e-4	42, 123, 999	0.679	0.659–0.694	0.63	Retained as a regularised three-seed configuration rather than for peak AUC.
Data augmentation	Rotation / translation + Spatial-Dropout	mainly 123	0.691 / 0.670	—	0.63	No convincing improvement over the retained architecture.

5.2.2.2 Key Findings

5.2.2.2.1 Can a lightweight CNN learn useful information from the full mammogram? Lightweight CNNs provided weak discrimination from the full mammogram. Validation AUC ranged from approximately 0.52 to 0.55, while BCE remained above the prevalence-based null baseline. Increasing model

capacity reduced BCE from approximately 1.25 to 0.80 but did not produce a comparable improvement in discrimination.

5.2.2.2.2 Does transfer learning improve the global representation?

Similarly to the local branch, I applied a ResNet50 model to validate whether pretrained features from imagenet could improve the representation.

Applying the lessons learnt from the local branch, a MaxPool2D + Flatten architecture was prioritized. The validation AUC increased to 0.617 on average, displaying signs of a better ordering of probabilities. The ResNet50 architecture therefore captures predictive signal from the data. However, the BCE remains as high as for lightweight model (0.9 on the validation set), encouraging further architecture improvements.

5.2.2.2.3 Can a small task-specific adapter improve frozen ResNet50 features?

Following the approach used for the local branch, a Dense8 layer was introduced used as a small task-specific adapter. Its activation parameter is set as ‘relu’.

The naive Dense8 adapter substantially reduced BCE, but discrimination became highly seed-dependent, with validation AUC ranging from 0.491 to 0.656. This suggested that the adapter activation was unstable. Batch normalization was therefore introduced before the ReLU activation to stabilise the intermediate representation.

With this structured Dense–BatchNormalization–ReLU adapter, validation AUC became much more consistent across the two evaluated seeds, ranging from 0.681 to 0.685, while BCE decreased to approximately 0.63.

5.2.2.2.4 Can additional regularisation improve generalisation?

Additional regularisation produced only limited improvements in discrimination. Adding a second dropout layer increased the validation AUC to 0.693 for seed 123 while maintaining the validation BCE around 0.63. The L2-regularised configuration was thus evaluated across three seeds and retained because it maintained BCE around 0.63 with validation AUC ranging from 0.659 to 0.694. Regularisation therefore did not provide a clear increase in peak AUC, but helped retain a satisfactory AUC–BCE trade-off across repeated training runs.

5.3 Fusion Model

5.3.1 Predictive performance

To obtain a more robust estimate of generalisation, the local, global, symmetric-fusion and residual-fusion models were evaluated using five-fold patient-grouped cross-validation. Each patient appeared in exactly one outer evaluation fold. The same outer patient partitions were used for all architectures, enabling paired comparison on matched observations.

Table 9: Mean $\pm$ standard deviation across five patient-grouped outer folds.

Architecture	ROC-AUC	BCE	AP	Brier
Global	0.651 ± 0.073	0.679 ± 0.043	0.643 ± 0.084	0.240 ± 0.019
Local	0.692 ± 0.106	0.632 ± 0.049	0.694 ± 0.121	0.222 ± 0.023
Symmetric fusion	0.723 ± 0.092	0.623 ± 0.080	0.728 ± 0.100	0.216 ± 0.034
Residual fusion	0.734 ± 0.076	0.602 ± 0.056	0.742 ± 0.093	0.208 ± 0.025

Across the five folds, residual fusion achieved the highest mean ROC-AUC and AP and the lowest BCE and Brier score. However, performance varied substantially between patient partitions, particularly for the local and symmetric models.

Pooled out-of-fold evaluation produced a similar ordering. Residual fusion achieved ROC-AUC 0.733, AP 0.745, BCE 0.602 and Brier score 0.208. Its Brier Skill Score relative to the prevalence-only baseline was 0.166, compared with 0.137 for symmetric fusion, 0.113 for the local branch and 0.039 for the global branch.

5.3.1.1 Does fusion improve prediction over the individual branches?

The local branch remained the strongest individual model, with pooled OOF ROC-AUC of 0.697 compared with 0.646 for the global branch. Symmetric fusion increased pooled ROC-AUC to 0.724, but its advantage over the local branch was not consistent under patient-level bootstrap resampling.

For symmetric fusion minus local, the 95% bootstrap confidence intervals included zero for all four metrics: ROC-AUC $+0.027$ $[-0.004, 0.057]$, BCE -0.009 $[-0.037, 0.020]$, AP $+0.005$ $[-0.035, 0.046]$, and Brier -0.006 $[-0.018, 0.006]$. The symmetric architecture therefore produced numerically favourable aggregate results but did not provide clear evidence of a systematic improvement over the local model.

Residual fusion produced a different result. Relative to the local branch, the paired OOF differences were:

Metric	Residual $-$ Local	95% bootstrap CI
ROC-AUC	$+0.036$	$[+0.012, +0.060]$
BCE	-0.030	$[-0.048, -0.011]$
AP	$+0.038$	$[+0.016, +0.059]$
Brier	-0.013	$[-0.022, -0.005]$

All four intervals excluded zero in the favourable direction. This provides evidence that combining the local prediction with contextual information from the full mammogram improved both discrimination and probabilistic prediction relative to the local branch alone.

5.3.1.2 Should local and global representations contribute symmetrically? Residual fusion achieved only a small pooled ROC-AUC improvement over symmetric fusion, 0.733 versus 0.724.

The paired bootstrap difference in ROC-AUC was +0.009, with a 95% confidence interval of [-0.008, 0.026]. The interval included zero, so the analysis did not provide clear evidence that residual fusion systematically improved ROC ranking relative to symmetric fusion.

The differences were clearer for BCE and average precision. Residual fusion reduced BCE by 0.021 [95% CI: -0.039, -0.003] and increased AP by 0.032 [0.005, 0.059]. Its Brier score was also lower by 0.007, although the confidence interval narrowly included zero [-0.014, 0.0001].

These results suggest that the main advantage of residual fusion was not a large increase in ROC-AUC, but a more effective use of contextual information for probabilistic prediction and precision–recall performance.

5.3.1.3 How stable were the models across patient partitions? Substantial between-fold variation was observed. The final outer fold was particularly difficult. Local ROC-AUC fell to 0.520, while the global branch achieved approximately 0.641. Symmetric and residual fusion achieved 0.573 and 0.609 respectively. Residual fusion therefore did not eliminate the effect of the difficult partition, although it reduced the degradation observed with symmetric fusion.

This variation indicates that model performance is sensitive to patient composition and that a single fixed validation/test partition would provide an incomplete estimate of generalisation stability.

5.3.2 Calibration

Calibration was assessed from the pooled OOF predictions using reliability diagrams and expected calibration error (ECE).

Architecture	ECE
Global	0.0797
Local	0.0589
Symmetric fusion	0.0540
Residual fusion	0.0240

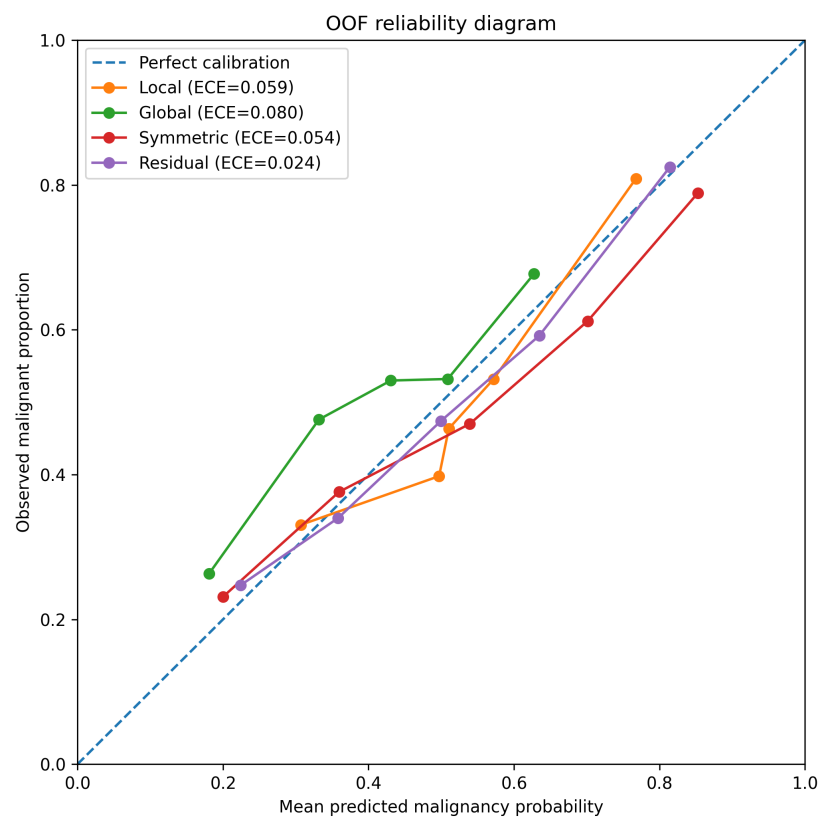


Figure 12: Model calibration comparison across models

Residual fusion produced the lowest descriptive ECE. Together with its lower BCE and Brier score, this indicates that its improvement was not limited to discrimination. However, ECE depends on the calibration-bin definition and was therefore interpreted descriptively rather than as a formal statistical comparison.

5.3.3 Operating threshold

The proposed threshold should maintain a high recall, because the risk of false negative offsets the risk of false positive in a medical environment. I thus defined the operating threshold to a minimum sensitivity of 85%, measured on the validation set as displayed in hereunder graph.

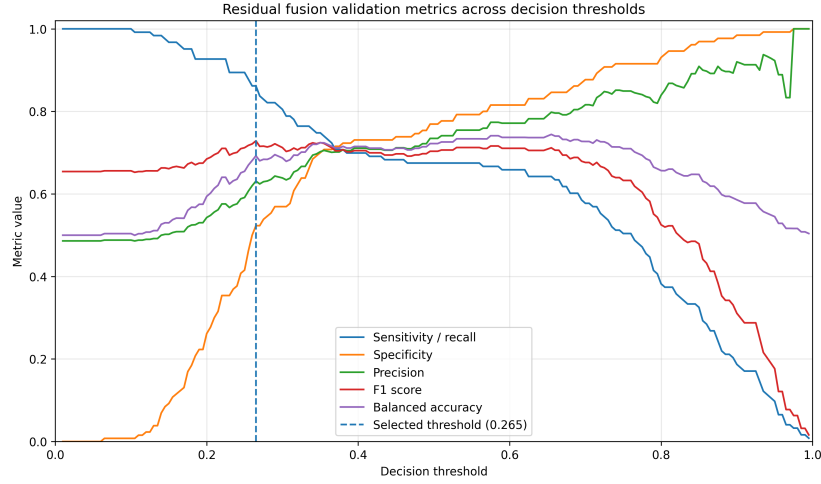


Figure 13: Threshold definition

On the validation set, the threshold maximising balanced accuracy was approximately 0.64. However, the retained operating threshold was 0.265 because it was the validation-selected threshold satisfying the predefined minimum sensitivity of 85%. This threshold was then applied unchanged to the test set.

Table 12: Test-set performance at the retained operating threshold of 0.265.

Metric	Test result
Threshold	0.265
Sensitivity / Recall	0.884
Specificity	0.537
Precision	0.563
F1-score	0.688
Balanced accuracy	0.711

Metric	Test result
Accuracy	0.677
True positives (TP)	130
True negatives (TN)	117
False positives (FP)	101
False negatives (FN)	17

On the paired test set, the retained threshold achieved a sensitivity of 88.4% and a specificity of 53.7%. The model therefore maintained the targeted high sensitivity on unseen data, at the cost of 101 false positives and 17 false negatives.

On this independent test set, the model achieved a ROC-AUC of 0.813, an average precision of 0.746, a binary cross-entropy of 0.543, and a Brier score of 0.179.

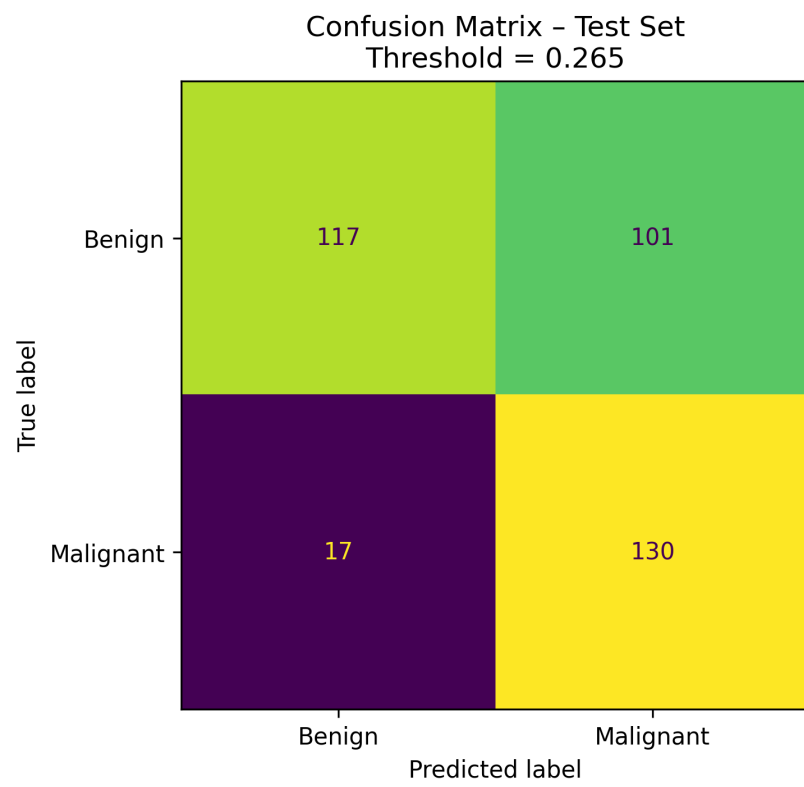


Figure 14: Confusion matrix

5.3.4 Interpretability

5.3.4.1 Grad-CAM methodology I implemented a grad-cam to compare the areas reviewed by the model with the actual ROI. Results are mixed.

The ROI is outlined with a purple line. The areas reviewed by the model are highlighted in tones edging to red. The model does focus on the breast and leaves the background aside in a large majority of spot checks.

Grad-CAM analysis showed that the spatial interpretation depended strongly on the class being explained. In one malignant case correctly classified at the selected operating threshold, the local branch produced a malignancy probability of 0.367, while the contextual correction reduced the final probability to 0.331. When malignant evidence was visualised, almost no Grad-CAM energy overlapped the annotated ROI (enrichment = 0.016). Conversely, the Grad-CAM generated for the benign class showed approximately twice as much activation within the ROI as expected from its area alone (enrichment = 2.00). This result suggests that the contextual branch can associate lesion or surrounding features with evidence for the competing class and may therefore weaken an otherwise correct local prediction. It also demonstrates that Grad-CAM localisation must be interpreted with respect to the selected target class rather than as a class-independent localisation map.

Grad-CAM vs lesion ROI | True: MALIGNANT | Pred: MALIGNANT | Target: MALIGNANT | P(malignant): 0.3307 | Correction: -0.1619

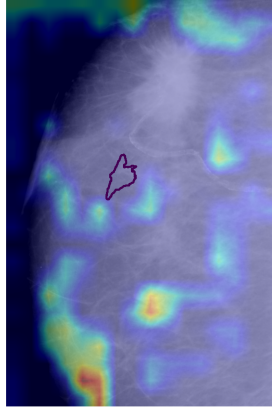


Figure 15: Global Grad-CAM for malignant evidence

5.3.4.2 Case analysis Although the global model orients the classification towards the wrong decision, it does not contribute to a change of classification because the local branch confidence remains above the selected operating threshold.

In hereunder example, the global branch heavily contributes to a change of decision towards the correct classification (benign instead of malignant). The

Grad-CAM vs lesion ROI | True: MALIGNANT | Pred: MALIGNANT | Target: BENIGN | P(malignant): 0.3307 | Correction: -0.1619

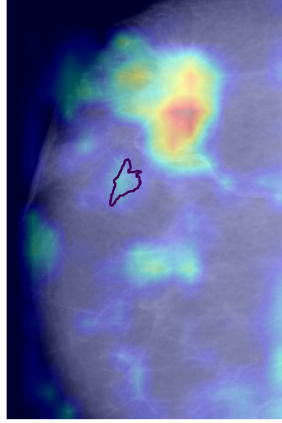


Figure 16: Global Grad-CAM for benign evidence on the same case

contextual branch produces a strong negative logit correction, indicating that its net contribution favours the benign class. The local model predicts malignancy with a probability 0.397 (above 0.265). The global context pushes it to 0.25.

Grad-CAM vs lesion ROI | True: BENIGN | Pred: BENIGN | Target: MALIGNANT | P(malignant): 0.2500 | Correction: -0.6796

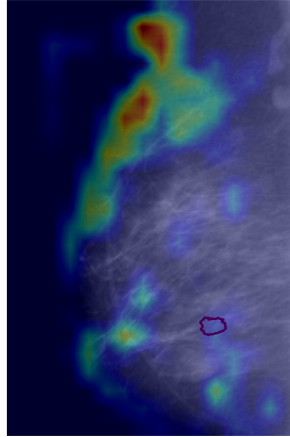


Figure 17: Contextual correction of a local false-positive prediction

6 Conclusion

This project investigated whether global mammographic context can improve the classification of a known breast mass when combined with an ROI-centred lesion representation. It was motivated by the global-local principle demonstrated

by architectures such as GMIC, which showed that information extracted from the full mammogram can complement detailed local representations (Shen et al. 2021). The present work examined whether this principle remains beneficial in a different setting, where lesion localisation is already provided by expert annotations and the task is restricted to benign–malignant CADx classification.

The results support this principle, but also qualify it. The local branch was the strongest standalone representation overall, confirming that lesion-centred morphology provides the dominant predictive signal for this task. Across patient-grouped cross-validation, the local model achieved a pooled OOF ROC-AUC of 0.697, compared with 0.646 for the global branch. Global mammographic information was therefore substantially weaker when used independently.

However, the weaker global representation was not necessarily uninformative. Its usefulness depended strongly on how it was incorporated. Symmetric fusion increased pooled ROC-AUC to 0.724, but patient-level bootstrap intervals did not provide clear evidence of systematic improvement over the local model. In contrast, residual fusion achieved the strongest overall performance, with pooled OOF ROC-AUC of 0.733, AP of 0.745, BCE of 0.602 and Brier score of 0.208. Relative to the local branch, residual fusion improved ROC-AUC by 0.036, AP by 0.038, and reduced BCE by 0.030 and Brier score by 0.013; the 95% patient-level bootstrap confidence intervals excluded zero for all four paired differences.

The comparison between the two fusion strategies provides the main architectural insight of this study. Residual fusion did not demonstrate a clear ROC-AUC advantage over symmetric fusion, since the bootstrap interval for their AUC difference included zero. Nevertheless, it achieved lower BCE and higher AP, while also producing the lowest descriptive ECE (0.024). These findings suggest that, when lesion information is already strongly predictive, global context may be more useful as a constrained correction to the local decision than as an equally weighted competing representation.

This offers a complementary perspective on GMIC. Rather than contradicting its global-local principle, the results suggest that the optimal relationship between global and local information depends on the task. In the present CADx setting, explicit lesion localisation reduced the need for the global branch to discover suspicious regions. Preserving the lesion-based prediction as a baseline and allowing mammographic context to modify it therefore provided a more effective formulation.

The study also identified important limitations. Performance varied substantially between patient partitions, showing that a single fixed split would have provided an incomplete estimate of generalisation. CBIS-DDSM remains relatively small, only mass lesions were considered, and the clinical meaning of the contextual features responsible for individual residual corrections remains uncertain. Grad-CAM analysis further showed that global activations were not systematically concentrated within the annotated lesion, so improved predictive performance alone cannot establish that clinically meaningful contextual

information was learned.

Overall, this study supports the value of global-local modelling for mammographic CADx while showing that how context is incorporated may matter as much as whether it is incorporated at all.

Appendix

Appendix 1

```
for dcm_abs_path in abs_folder.rglob("*.dcm"):
    ds = pydicom.dcmread(dcm_abs_path, stop_before_pixels=True)
    series_description = str(
        getattr(ds, "SeriesDescription", "")
    ).strip()

    if series_description == "type_of_file":
        return str(dcm_abs_path.relative_to(images_root))

return None
```

Appendix 2

```
splitter = StratifiedGroupKFold(
    n_splits=5,
    shuffle=True,
    random_state=trial_seed,
)

for train_indices, val_indices in splitter.split(
    dummy_x,
    labels,
    groups,
):
    train_rate = development_df.iloc[train_indices]["label"].mean()
    val_rate = development_df.iloc[val_indices]["label"].mean()

    prevalence_error = max(
        abs(train_rate - overall_malignant_rate),
        abs(val_rate - overall_malignant_rate),
    )

    size_error = abs(
        len(val_indices) / len(development_df) - val_prc
    )

    score = prevalence_error + size_error
```

Appendix 3

```
def orient_by_breast_mass(image, mask=None, target_side="right", threshold=0.05):
```

```

image_np = image.numpy() if isinstance(image, tf.Tensor) else np.asarray(image)

image_2d = np.squeeze(image_np)

h, w = image_2d.shape

tissue = image_2d > threshold

left_mass = tissue[:, : w // 2].sum()
right_mass = tissue[:, w // 2 :].sum()

breast_is_left = left_mass > right_mass

if target_side == "right":
    should_flip = breast_is_left
elif target_side == "left":
    should_flip = not breast_is_left

if should_flip:
    image = np.asarray(image)
    image = np.fliplr(image)

    if mask is not None:
        mask = np.asarray(mask)
        mask = np.fliplr(mask)

if mask is not None:
    return image, mask

return image

```

Appendix 4

```

for filters in [32, 64, 128]:
    x = tf.keras.layers.Conv2D(
        filters,
        kernel_size=3,
        padding="same"
    )(x)
    x = tf.keras.layers.ReLU()(x)
    x = tf.keras.layers.MaxPooling2D(pool_size=2)(x)

x = tf.keras.layers.Flatten()(x)

```

Appendix 5

```
def build_resnet50_transfer(
    input_shape=(PIXELS_H, PIXELS_W, 1),
    dropout_rate=0.5,
):
    inputs = tf.keras.Input(
        shape=input_shape,
        name="mammogram_input",
    )

    # input passed 3 times
    x = tf.keras.layers.Concatenate(
        axis=-1,
        name="grayscale_to_rgb",
    )([inputs, inputs, inputs])

    x = tf.keras.layers.Rescaling(
        scale=255.0,
        name="restore_255_scale",
    )(x)

    x = tf.keras.applications.resnet50.preprocess_input(x)

    base_model = tf.keras.applications.ResNet50(
        include_top=False,
        weights="imagenet",
        input_shape=(input_shape[0], input_shape[1], 3),
    )

    base_model.trainable = False

    x = base_model(x, training=False)

    x = tf.keras.layers.Flatten(
        name="resnet_global_flatten",
    )(x)

    x = tf.keras.layers.Dropout(
        0.5,
        name="classification_dropout",
    )(x)

    outputs = tf.keras.layers.Dense(
        units=1,
        activation="sigmoid",
```

```

        name="classification_output",
    )(x)

    model = tf.keras.Model(
        inputs=inputs,
        outputs=outputs,
        name="local_resnet50_transfer",
    )

    return model, base_model

```

Appendix 6

```

x = tf.keras.layers.MaxPooling2D(
    pool_size=(4, 4),
    strides=(4, 4),
    padding="same",
    name="resnet_channel_max_pooling",
)(x)

# Flatten layer here

x = tf.keras.layers.Dense(
    units=16,
    activation="relu",
    name="mammography_adapter",
)(x)

# Dropout layer here

```

Appendix 7

```

# Solve dying ReLU issue
x = tf.keras.layers.Dense(
    units=8,
    activation=None,
    use_bias=False,
    name="mammography_adapter",
)(x)

x = tf.keras.layers.BatchNormalization(
    name="mammography_adapter_batch_norm"
)(x)

x = tf.keras.layers.ReLU(
    name="mammography_adapter_relu",

```

```
) (x)
```

Appendix 8

The feature layers are retrieved as follow.

```
local_embedding_layer = local_model.get_layer(local_embedding_layer_name)
global_embedding_layer = global_model.get_layer(global_embedding_layer_name)

local_extractor = tf.keras.Model(
    inputs=local_model.input,
    outputs=local_embedding_layer.output,
    name="local_feature_extractor",
)

global_extractor = tf.keras.Model(
    inputs=global_model.input,
    outputs=global_embedding_layer.output,
    name="global_feature_extractor",
)
```

The backbone is then frozen.

```
local_extractor.trainable = False
global_extractor.trainable = False
```

The embeddings are retrieved using respective model dummy inputs.

```
local_input = tf.keras.Input(
    shape=local_model.input_shape[1:],
    name="local_input",
)

global_input = tf.keras.Input(
    shape=global_model.input_shape[1:],
    name="global_input",
)

local_embedding = local_extractor(
    local_input,
    training=False,
)

global_embedding = global_extractor(
    global_input,
    training=False,
)
```


Embeddings are eventually concatenated before having a Dense layer applied, to generate unique trainable parameters.

```
fused_features = tf.keras.layers.Concatenate(
    name="feature_fusion",
)(
    [
        local_embedding,
        global_embedding,
    ]
)
```

Appendix 9

On one hand, the local branch's logits are retrieved with this architecture.

```
local_logit_layer = tf.keras.layers.Dense(
    units=1,
    activation=None,
    use_bias=local_output_layer.use_bias,
    trainable=False,
    name="frozen_local_baseline_logit",
)

local_logit = local_logit_layer(
    local_classifier_input
)

local_logit_layer.set_weights(
    local_output_layer.get_weights()
)
```

The global branch has some local context added with hereunder standard embedding fusion.

```
fused_features = tf.keras.layers.Concatenate(
    name="residual_feature_fusion",
)(
    [
        local_embedding,
        global_embedding,
    ]
)
```

The concatenated embedding has two Dense layer applied. The first one enriches the parameters. The second one return a Dense1 that can be added to the local logit branch.

```
correction = tf.keras.layers.Dense(
```

```

        units=correction_units,
        activation=None,
        use_bias=False,
        kernel_regularizer=regularizer,
        name="contextual_correction_adapter",
    )(fused_features)

    # Layer Normalization

    # ReLU layer

    # Dropout layer

    contextual_logit_correction = tf.keras.layers.Dense(
        units=1,
        activation=None,
        kernel_initializer="zeros",
        bias_initializer="zeros",
        kernel_regularizer=regularizer,
        name="contextual_logit_correction",
    )(correction)

    final_logit = tf.keras.layers.Add(
        name="local_logit_plus_contextual_correction",
    )(
        [
            local_logit,
            contextual_logit_correction,
        ]
    )

```

Appendix 10

```

outer_cv = StratifiedGroupKFold(
    n_splits=N_OUTER_FOLDS,
    shuffle=True,
    random_state=TRAINING_SEED,
)

for outer_fold, (outer_train_idx, outer_eval_idx) in enumerate(
    outer_cv.split(
        dev_df,
        y=dev_df["label"],
        groups=dev_df["patient_id"],
    )
):

```

```

outer_train_df = dev_df.iloc[outer_train_idx].copy()
outer_eval_df = dev_df.iloc[outer_eval_idx].copy()

inner_cv = StratifiedGroupKFold(
    n_splits=5,
    shuffle=True,
    random_state=TRAINING_SEED + outer_fold,
)

inner_train_idx, inner_val_idx = next(
    inner_cv.split(
        outer_train_df,
        y=outer_train_df["label"],
        groups=outer_train_df["patient_id"],
    )
)

```

Appendix 11

```

def threshold_grid_search(
    y_true: np.ndarray,
    y_probability: np.ndarray,
    thresholds: np.ndarray | None = None,
) -> pd.DataFrame:

    y_true = np.asarray(
        y_true,
        dtype=np.int32,
    ).reshape(-1)

    y_probability = np.asarray(
        y_probability,
        dtype=np.float64,
    ).reshape(-1)

    rows: list[dict[str, Any]] = []

    for threshold in thresholds:
        y_predicted = (
            y_probability >= threshold
        ).astype(np.int32)

        tn, fp, fn, tp = confusion_matrix(
            y_true,
            y_predicted,
            labels=[0, 1],

```

```

).ravel()

# Metric calculations using tn, fp, fn, tp

# return results

```

Appendix 12

```

with tf.GradientTape() as tape:
    # Final convolutional feature maps from both frozen ResNet50 branches
    local_feature_maps = local_feature_model(
        local_image,
        training=False,
    )
    global_feature_maps = global_feature_model(
        global_image,
        training=False,
    )

    tape.watch(local_feature_maps)
    tape.watch(global_feature_maps)

    # Reconstruct local and global embeddings
    local_embedding = apply_layers_inference(
        local_feature_maps,
        local_head_to_embedding,
    )

    local_classifier_input = apply_layers_inference(
        local_embedding,
        local_embedding_to_classifier_input,
    )

    local_logit = call_layer_inference(
        local_baseline_logit_layer,
        local_classifier_input,
    )

    global_embedding = apply_layers_inference(
        global_feature_maps,
        global_head_to_embedding,
    )

    # Reconstruct residual contextual correction
    local_embedding = call_layer_inference(
        local_normalization_layer,

```

```

        local_embedding,
    )
    global_embedding = call_layer_inference(
        global_normalization_layer,
        global_embedding,
    )

    fused_features = fusion_layer(
        [local_embedding, global_embedding]
    )

    correction = call_layer_inference(
        correction_adapter_layer,
        fused_features,
    )
    correction = call_layer_inference(
        correction_normalization_layer,
        correction,
    )
    correction = call_layer_inference(
        correction_relu_layer,
        correction,
    )
    correction = call_layer_inference(
        correction_dropout_layer,
        correction,
    )

    contextual_logit_correction = call_layer_inference(
        correction_output_layer,
        correction,
    )

    final_logit = final_logit_layer(
        [local_logit, contextual_logit_correction]
    )

    # Explain malignant (+logit) or benign (-logit) evidence
    target_score = (
        final_logit[:, 0]
        if target_class == 1
        else -final_logit[:, 0]
    )

    local_gradients, global_gradients = tape.gradient(
        target_score,

```

```
        [local_feature_maps, global_feature_maps],  
    )  
  
    local_heatmap = make_gradcam_heatmap(  
        local_feature_maps,  
        local_gradients,  
    )  
  
    global_heatmap = make_gradcam_heatmap(  
        global_feature_maps,  
        global_gradients,  
    )
```

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